University of Asia Pacific

Department of Computer Science and Engineering

FINAL PROJECT REPORT

CSE 438: Robotics Lab

Submitted to

Md. Rasheduzzaman

Department of CSE, University of Asia Pacific

Team Name: FireX

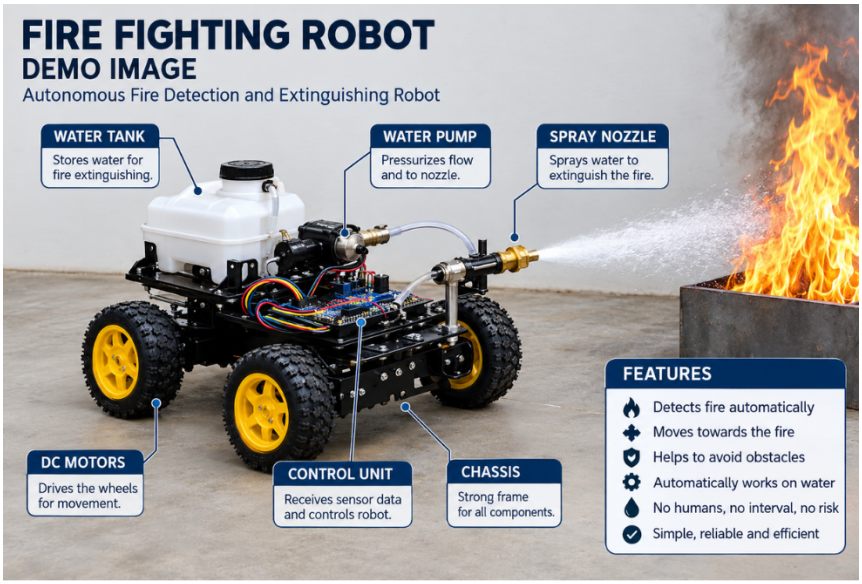
Date: 03/11/2026

Team Members

Name	Registration ID
Joynove Binta Mim	23101179
Shahriar Hossain Saad	23101192
Maharun Nesa Jarin	23101195
Hafiz Al Asad Tamim	23101197

FireX: Fire Fighting Robot

An Autonomous Fire Detection and Extinguishing System



1. Abstract

Fire accidents are a major safety concern in Bangladesh, particularly in factories, workshops, and industrial areas. Firefighters often face extreme heat, smoke, and other dangerous conditions while responding to fire emergencies. To address this problem, this project proposes FireX, a low-cost autonomous firefighting robot designed to detect and extinguish small fires while reducing human exposure to danger.

The main mission of FireX is to develop an affordable robot that can detect fire, identify its direction, avoid obstacles, and activate a fire-extinguishing system with minimal human intervention. Its vision is to promote safer firefighting practices through accessible robotic technology for workshops, factories, and other high-risk environments. FireX is intended for small, controlled fires and is not designed to replace professional firefighters or handle large industrial and chemical fires. The project demonstrates how low-cost robotics can support initial fire response and help improve safety in hazardous environments.

2. Introduction

FireX is a low-cost autonomous firefighting robot designed to detect fire and identify its direction using IR flame sensors. It can move automatically toward the fire while avoiding obstacles with the help of an ultrasonic sensor. During fire detection, it provides an instant warning and activates a water pump to spray water toward the fire using a servo-controlled nozzle. The robot aims to reduce human exposure to dangerous fire environments and provides an affordable firefighting solution using locally available components.

3. Objectives

3.1 General Objective

To design and develop FireX, a low-cost autonomous firefighting robot that helps improve fire safety and reduce human exposure to dangerous fire environments.

3.2 Specific Objectives

1. To detect fire and identify its direction using IR flame sensors.
2. To detect and avoid obstacles, using an ultrasonic sensor.
3. To navigate autonomously toward the detected fire.
4. To activate a water pump and spray water to extinguish small fires.
5. To control robot movement and the extinguishing mechanism using an Arduino microcontroller.

4. Proposed Methodology

The FireX project will be developed by integrating an Arduino microcontroller with IR flame sensors, an ultrasonic sensor, DC motors, a servo motor, and a mini water pump. The flame sensors will detect fire and identify its approximate direction, while the ultrasonic sensor will help the robot detect and avoid obstacles. Based on sensor readings, the Arduino will control the robot's movement toward the fire. When the robot reaches a suitable distance, it will stop, aim the nozzle toward the fire using the servo motor, and activate the water pump to spray water. Finally, the system will be tested under controlled conditions to evaluate its fire detection, navigation, obstacle avoidance, and extinguishing performance.

4.1 System Design and Planning

The FireX system is designed using an Arduino microcontroller as the main control unit. The design includes IR flame sensors for fire detection and direction identification, an ultrasonic sensor for obstacle avoidance, and DC motors with a motor driver for robot movement. A servo motor will

control the direction of the water nozzle, while a mini water pump will spray water to extinguish small fires. A suitable battery will supply power to the system. The planning includes hardware assembly, Arduino programming, sensor integration, and controlled testing to ensure that FireX can detect fire, navigate toward it, avoid obstacles, and activate the extinguishing mechanism within the estimated budget of BDT 3,000–4,000.

4.2 Hardware Development

The hardware components listed in the report are:

- Arduino Uno / Nano
- IR Flame Sensor
- MQ2 (Gas sensor)
- Ultrasonic Sensor (HC-SR04), SG90
- Servo Motor
- DC Gear Motors, L298N
- Motor Driver
- Water Pump (5V)
- Tubing / Pipe
- Battery (9V/12V)
- Jumper Wires & Breadboard

Hardware Assembly Process

1. **Sensors:** Three IR flame sensors (left, right, center) are placed on the front to detect direction.
2. **Motion:** An L298N motor driver controls four DC gear motors (wheels).
3. **Extinguishing:** A 5V mini water pump and a servo motor aim the nozzle toward the fire.

5. Software Development

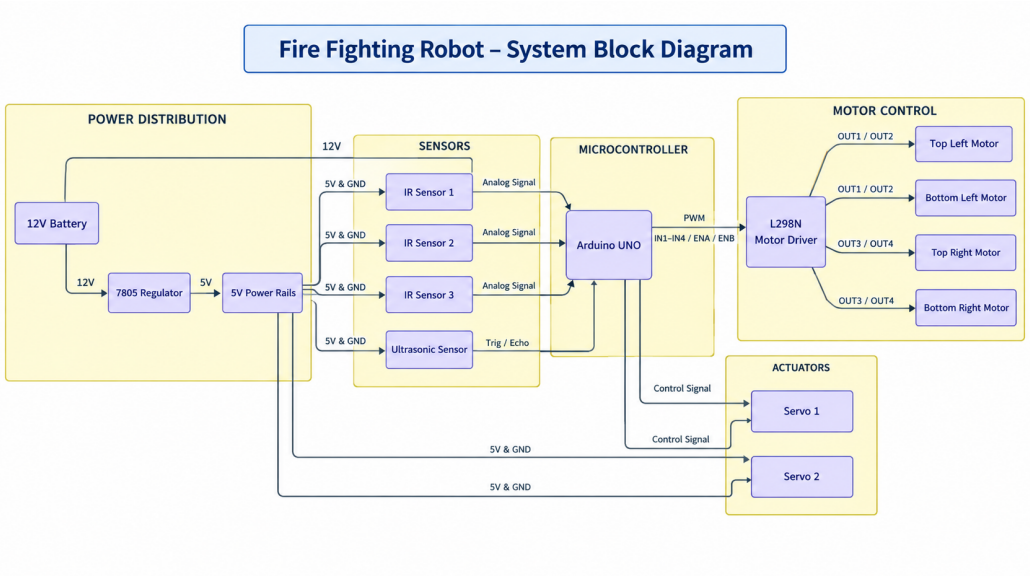


Figure 1: Overall Workflow of the Fire-X Robot System

5.1 Pseudocode

The following pseudocode describes the operational logic of the robot system. The complete Arduino program is provided in Section 8.

5.2 System Integration

- The 12V battery supplies power to the system.
- The 7805 regulator converts 12V into 5V for the sensors and other components.
- Three IR sensors detect the fire and identify its direction.
- The ultrasonic sensor detects obstacles in front of the robot.
- The Arduino UNO receives sensor signals and processes the data.
- The Arduino controls the L298N motor driver to move the robot.
- Four DC motors control the robot's movement.
- Two servo motors control the nozzle direction.
- When fire is detected, the robot moves toward the fire and stops at a suitable distance.
- Finally, the water pump turns on to spray water and extinguish the fire.

6. Budget

6.1 Budget Table

Component	Qty	Approx. Price (BDT)
Arduino Uno / Nano	1	500–800
IR Flame Sensor	3	150–300
Ultrasonic Sensor (HC-SR04)	1	150–200
SG90 Servo Motor	1	150–200
DC Geared Motors	4	300–500
L298N Motor Driver	1	200–300
Water Pump (5V)	1	150–250
Tubing / Pipe	1 set	100–150
Battery (9V/12V) or Power Bank	1	400–600
Jumper Wires & Breadboard	1 set	100–150

Figure 2: Original Budget Table

Estimated Total: Approx. BDT 3,000–4,000

7. Circuit Diagram

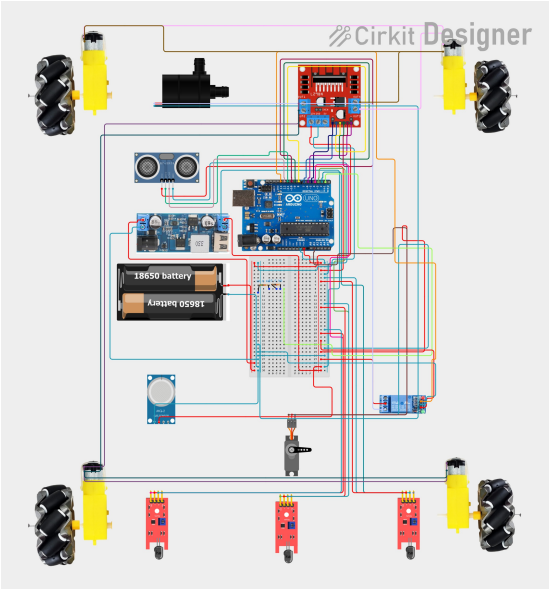


Figure 3: Circuit Diagram

8. Pseudocode

```
#include <Servo.h>

// ================= FLAME SENSORS =================
// Sensor output LOW = Fire detected
#define FLAME_CENTER 2
#define FLAME_LEFT 3
#define FLAME_RIGHT 4

// ================= MQ-2 GAS SENSOR =================
#define GAS_SENSOR A1

// ================= L293D MOTOR DRIVER =================
#define ENA 6
#define IN1 7
#define IN2 9

#define ENB 5
#define IN3 11
#define IN4 12

// ================= OTHER COMPONENTS =================
#define BUZZER 8
#define RELAY_PIN A0
#define SERVO_PIN 13

// Active LOW relay
#define PUMP_ON LOW
#define PUMP_OFF HIGH

// ================= SETTINGS =================
#define MOTOR_SPEED 120
#define SLOW_SPEED 90
#define APPROACH_TIME 250
#define SPRAY_TIME 5000

Servo waterServo;

// ================= SENSOR FUNCTIONS =================
bool centerFire() {
  return digitalRead(FLAME_CENTER) == LOW;
}

bool leftFire() {
  return digitalRead(FLAME_LEFT) == LOW;
}

bool rightFire() {
  return digitalRead(FLAME_RIGHT) == LOW;
}

bool anyFire() {
  return centerFire() || leftFire() || rightFire();
}

bool gasDetected() {
  return digitalRead(GAS_SENSOR) == LOW;
}

// ================= MOTOR CONTROL =================
void moveForward() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
  analogWrite(ENA, 180);
  analogWrite(ENB, 180);
}

void forwardLeft() {
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, HIGH);
  digitalWrite(IN4, LOW);
```

```
  analogWrite(ENA, 180);
  analogWrite(ENB, 180);
}

void forwardRight() {
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, HIGH);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, HIGH);
  analogWrite(ENA, 180);
  analogWrite(ENB, 180);
}

void stopMotors() {
  analogWrite(ENA, 0);
  analogWrite(ENB, 0);
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  digitalWrite(IN3, LOW);
  digitalWrite(IN4, LOW);
}

// ================= PUMP CONTROL =================
void pumpOn()  { digitalWrite(RELAY_PIN, PUMP_ON);  }
void pumpOff() { digitalWrite(RELAY_PIN, PUMP_OFF); }

// ================= SAFE STOP =================
void emergencyStop() {
  stopMotors();
  pumpOff();
  digitalWrite(BUZZER, LOW);
}

// ================= APPROACH FIRE =================
void approachFire() {
  unsigned long startTime = millis();
  while (millis() - startTime < APPROACH_TIME) {
    if (gasDetected() || !anyFire()) {
      stopMotors();
      return;
    }

    bool C = centerFire();
    bool L = leftFire();
    bool R = rightFire();

    if (C) {
      moveForward();
    }
    else if (L && !R) {
      forwardLeft();
    }
    else if (R && !L) {
      forwardRight();
    }
    else if (L && R) {
      moveForward();
    }
    delay(20);
  }
  stopMotors();
}

// ================= EXTINGUISH FIRE =================
void extinguishFire() {
  stopMotors();
  unsigned long startTime = millis();

  while (millis() - startTime < SPRAY_TIME) {
    if (gasDetected() || !anyFire()) break;

    pumpOn();
    waterServo.write(50);
    delay(250);

```

```
    if (gasDetected() || !anyFire()) break;
    waterServo.write(90);
    delay(250);

    if (gasDetected() || !anyFire()) break;
    waterServo.write(130);
    delay(250);

    if (gasDetected() || !anyFire()) break;
    waterServo.write(90);
    delay(250);
  }

  pumpOff();
  waterServo.write(90);
  stopMotors();
}

// ================= SETUP =================
void setup() {
  pinMode(FLAME_CENTER, INPUT_PULLUP);
  pinMode(FLAME_LEFT, INPUT_PULLUP);
  pinMode(FLAME_RIGHT, INPUT_PULLUP);
  pinMode(GAS_SENSOR, INPUT_PULLUP);

  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(ENB, OUTPUT);
  pinMode(IN3, OUTPUT);
  pinMode(IN4, OUTPUT);

  pinMode(BUZZER, OUTPUT);
  pinMode(RELAY_PIN, OUTPUT);

  digitalWrite(BUZZER, LOW);
  pumpOff();
  stopMotors();

  waterServo.attach(SERVO_PIN);
  waterServo.write(90);
  delay(1000);
}

// ================= MAIN LOOP =================
void loop() {
  if (gasDetected()) {
    emergencyStop();
    delay(200);
    return;
  }

  bool C = centerFire();
  bool L = leftFire();
  bool R = rightFire();

  if (!C && !L && !R) {
    emergencyStop();
    delay(100);
    return;
  }

  digitalWrite(BUZZER, HIGH);
  approachFire();
  stopMotors();
  digitalWrite(BUZZER, LOW);

  if (!gasDetected() && anyFire()) {
    extinguishFire();
  }

  stopMotors();
  pumpOff();
  delay(100);
}
```

Listing 1: FireX Fire Fighting Robot Arduino Program

8.1 Hardware Assembly

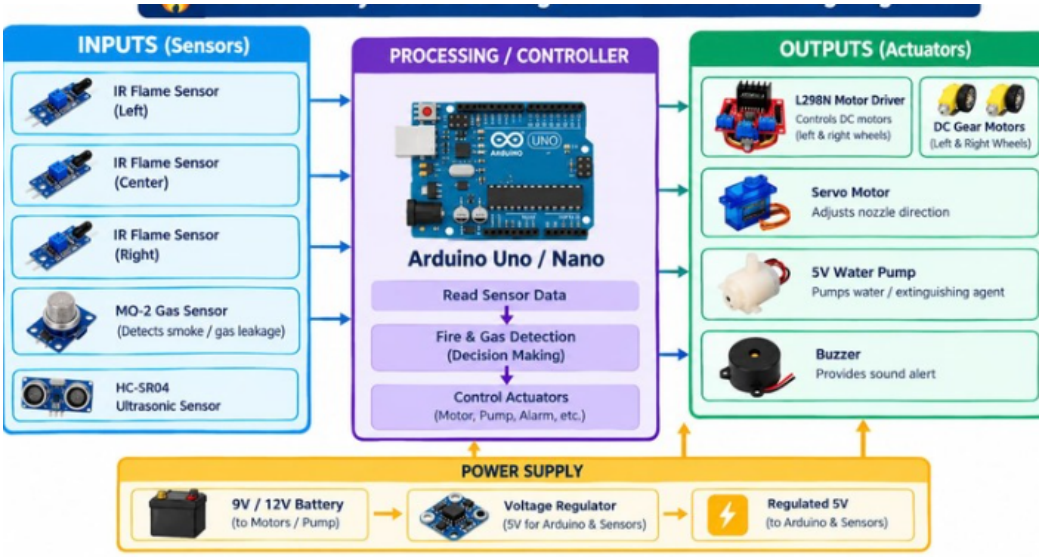


Figure 4: Main hardware of the system

8.2 Final Demonstration

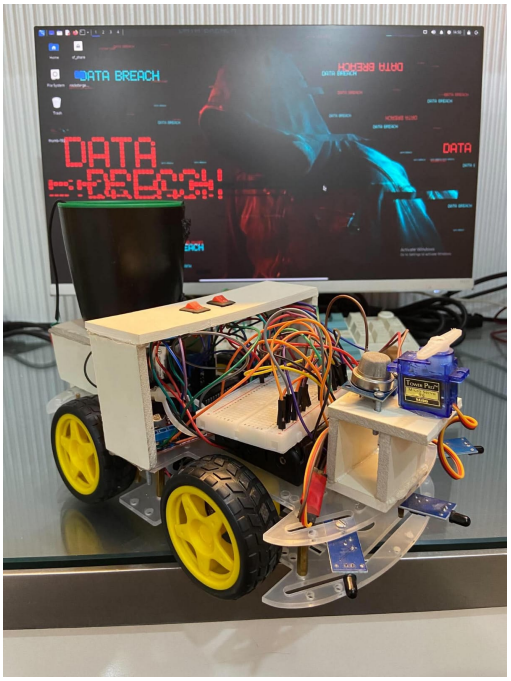


Figure 5: Fire-X Robot

The demonstrated system features and activities are:

- Fire detection and direction identification
- Autonomous robot movement
- Obstacle detection and avoidance
- Automatic water spraying

- Servo-controlled nozzle operation
- Complete system testing and performance evaluation

9. Gantt Chart

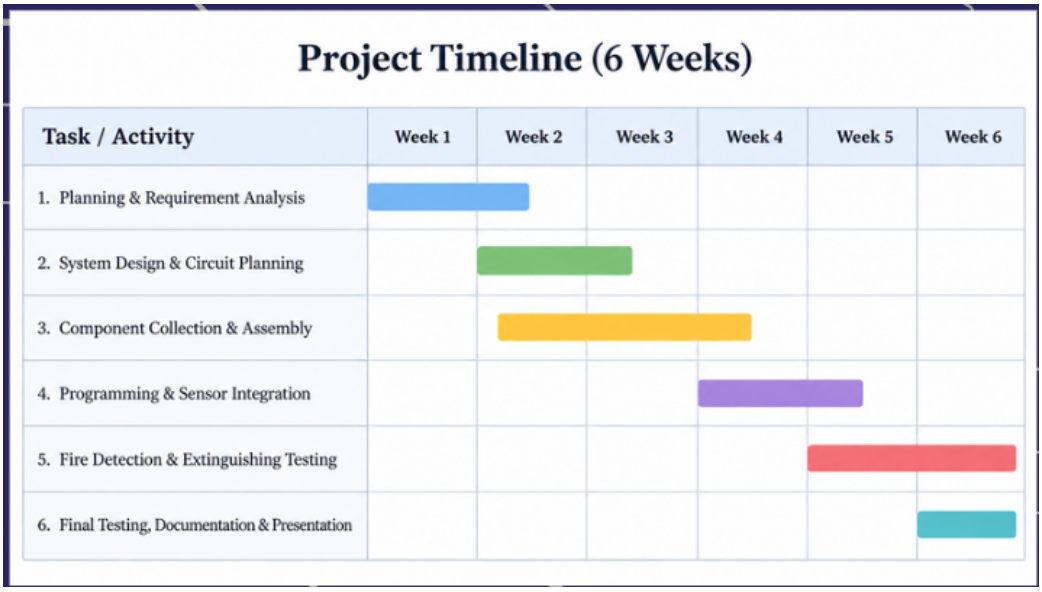


Figure 6: Project timeline

The project timeline is summarized in Table 1.

Task	Timeline	Short Description
1. Problem Identification and Planning	Week-1 to Week-2	The project problem was identified by analyzing excessive passive screen time among students and the lack of affordable robotics education. Initial planning, goal setting, and project roadmap preparation were completed during this phase.
2. System Design	Week-3	The overall architecture of the robot was designed, including hardware layout, software workflow, component selection, circuit planning, and interaction system design.
3. Budget & Material collection	Week-4	A low-cost budget plan was prepared and required materials such as Arduino, sensors, motors, batteries, and acrylic sheets were collected for implementation.
4. Demonstration of purchased component	Week-4	Fire detected by IR Flame sensor, Gas detected by MQ2 sensor, buzzer.
5. Hardware Assembly	Week-5 to Week-7	Complete robot chassis, setup IR Flame sensor, MQ2, and buzzer.
6. Software Development	Not specified in source	No timeline or short description was supplied for this row in the source report.
7. System Integration	Week-8 to Week-9	Robot detects fire, moves toward it, stops nearby, and activates the water pump to extinguish the fire.

8. Testing and Debugging	Week-10	All work perfectly. Errors were identified and corrected through debugging.
9. Performance Evaluation	Week-11 to Week-12	The overall system performance, educational effectiveness, usability, and interaction quality were evaluated through demonstrations and practical operation.
10. Report Writing	Week-13	Technical documentation including methodology, implementation, circuit design, survey findings, testing results, and analysis was compiled into the final professional report.
11. Final Presentation	Week-14	The completed project was demonstrated in robotics exhibitions and project showcases, where the system functionality, educational value, and innovation were presented to judges and participants.

Table 1: Project timeline description

10. Future Work

In the future, FireX can be improved by adding a camera, IoT-based remote monitoring, smoke and temperature sensors, AI-based fire detection, and advanced navigation systems. A larger battery, water tank, and stronger pump can also be used to increase operating time and firefighting capability. These improvements can make FireX more accurate, reliable, and useful in real-world environments.

11. References

1. An Autonomous Arduino-based Firefighting Robot for Laboratory Environments (2025).
2. Low-cost multi-sensing fire-fighting robot with obstacle avoidance mechanism.
3. A multi-sensor based fire fighting robot with wireless control and visual system (2020).

12. Conclusion

FireX is a highly relevant and practical project. It can be built on a low budget and directly addresses an urgent societal need. We are confident that within 6 weeks, we can produce a working version of this robot, which will pave the way for more advanced innovations in the future.

Survey evaluation, demonstrations, and exhibition participation confirmed the educational value and practical applicability of the project. The achievement of champion status in project showcases further validated the innovation and effectiveness of the proposed system. With future improvements and scalability, the project has strong potential to contribute to accessible robotics education and technological development in both rural and urban educational institutions.

13. Appendix

Appendix A: Survey Report Summary



Figure 7: Survey Report Picture

Appendix C: CEP Mapping

How Ks are addressed through the Project

Ks	Attributes	How Ks address through the project	CLOs	PLOs
K3	Engineering Fundamentals	Basic electronics and circuit fundamentals were applied: a 12V battery with a 7805 regulator for the 5V supply, an L298N motor driver controlling four DC motors, a relay for water pump switching, and digital/analog interfacing of the IR flame sensors and MQ-2 gas sensor.	CLO1	PO1, PO3
K4	Specialist Knowledge	Embedded systems and robotics knowledge: Arduino programming, sensor-based decision making (left/center/right flame sensors to determine fire direction), PWM motor speed control, servo-based nozzle sweeping, and emergency-stop logic when gas is detected.	CLO2	PO1, PO3
K5	Engineering Design	System block diagram (Figure-1), circuit design (Figure-3), hardware assembly and chassis design, all within a limited budget (BDT 3,000–4,000) and locally available components.	CLO3	PO3, PO11
K6	Engineering Practice	Practical work: circuit simulation in Cirkit Designer, component collection, assembly, testing, debugging, and demonstration in a controlled environment (Weeks 4–10).	CLO4	PO4, PO5

Table 2: K-attribute mapping

How As are addressed through the Project

As	Attributes	How As are Addressed through the Project

A1	Range of Resources	Several resource types were used: a 4-member team (people), a budget of BDT 3,000–4,000 (money), tools and equipment for assembly and testing, materials (acrylic chassis, tubing, wiring, batteries), and technologies (Arduino, IR flame sensors, MQ-2 gas sensor, L298N driver, relay, servo, water pump, Cirkit Designer).
A2	Level of Interaction	Hardware, software and power issues had to be handled together: sensor readings driving motor decisions, the pump and motors sharing a 12V supply and 5V regulator, and cost versus reliability and size versus water capacity. The team also had to coordinate hardware assembly, programming and testing.
A4	Consequences to Society and Environment	A fire robot has real safety consequences. It reduces firefighters' exposure to danger, but a failure could mean an unextinguished fire. The report limits FireX to small, controlled fires, and the gas-detection emergency stop, buzzer alert and automatic pump shut-off are safety measures. Water spraying has little environmental impact compared with chemical agents.

Table 3: A-attribute mapping

How Ps are addressed through the Project

Ps	**Attributes**	**How Ps are Addressed through the Project**
P1	Depth of Knowledge	The project requires K3 (electronics), K4 (embedded systems/robotics) and K5 (design) together. It cannot be solved with a single basic concept.
P2	Conflicting Requirements	Low cost (BDT 3,000–4,000) vs. reliable fire detection, battery life vs. motor and pump power, robot size vs. water capacity, and movement speed vs. accurate fire direction detection. These trade-offs had to be balanced.
P3	Depth of Analysis	The combination of center/left/right sensor readings was analyzed to decide between moving forward and turning. Approach time and spray time were tuned, the pump and motors shut off when gas is detected, and spraying stops automatically when the fire is gone.
P7	Interdependence	Sensors, Arduino, motor driver, relay/pump, servo, buzzer and power supply all depend on one another. Failure of one subsystem affects the whole robot's operation.

Table 4: P-attribute mapping